Article

Expert-Consistency-Guided Proposal Refinement for 3D Object Detection from Sparse Radar Point Clouds

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Abstract

Sparse radar point clouds provide limited evidence for 3D object detection, making weak proposals especially sensitive to ranking and suppression. Standard post-processing depends on the primary detector's scores and geometry and can therefore remove useful candidates. We introduce an expert-consistency-guided refinement framework that acts on detector outputs without changing the backbone. Candidate boxes are retained and reweighted according to agreement with an independently trained radar expert; supported boxes are then adjusted by local geometric voting. The framework is evaluated on Astyx, MAN TruckScenes, V2X-Radar-V, and K-Radar with three training seeds. At a 3D intersection-over-union threshold of 0.50, the strict route improves AP in all 12 matched dataset–seed comparisons, with a mean gain of 0.9726 AP. The high-performance route raises macro AP from 35.3552 to 39.1214, a gain of 3.7662 AP. Ablation experiments and fixed-parameter controls show that expert-based reweighting drives the consistent gain, while local voting further adjusts the geometry of supported boxes. The two routes therefore offer a practical choice between consistent improvements and higher average accuracy for 3D detection from sparse radar point clouds.

Keywords: 3D object detection; radar point cloud; autonomous driving; proposal refinement; expert consistency; geometric voting

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1. Introduction

Automotive perception must remain reliable when poor illumination, adverse weather, or difficult viewing geometry limits camera observations. Millimetre-wave radar is valuable in these conditions because it directly measures range and radial velocity. It also remains operational in situations that can degrade passive optical sensing. Recent 4D radar datasets cover a broader range of driving, weather, and cooperative-perception settings [25,29]. These developments have made radar-only three-dimensional (3D) object detection useful beyond its conventional role in camera–radar fusion. A radar-only detector can provide an independent source of spatial and motion evidence when visual information is unavailable or unreliable.

The same measurements that make radar useful also produce a difficult input for 3D detection. Radar point clouds contain far less geometric evidence than dense lidar scans, and an object may generate only a few irregular returns. Their number, amplitude, and Doppler distribution vary with distance, aspect, occlusion, sensor configuration, and preprocessing [36,37]. Reflections may cover only part of the visible object surface and need not coincide with its geometric centre. Return density also falls with range and can change markedly between measurement systems. A detector must therefore reconstruct a complete 3D box from partial, noisy, and measurement-dependent observations.

Existing 3D detectors handle irregular point sets through point-based, voxel-based, or pillar-based representations [1–5]. PointPillars is particularly suitable for controlled radar experiments because it converts occupied pillars into an efficient bird's-eye-view representation. Radar-specific work has also explored denser representations and multi-modal fusion to compensate for missing geometry [21,33,38]. These approaches improve the evidence available to the detector, but they do not remove the uncertainty of the final proposal set. Radar-only detection still has to decide which weak hypotheses are useful without relying on image semantics or dense lidar geometry.

This decision becomes important after the network has generated its candidate boxes. Several uncertain proposals may correspond to the same object, yet score filtering and non-maximum suppression (NMS) can remove them before their combined evidence is considered. Hard NMS favours a single high-scoring box, even when neighbouring boxes contain complementary localization information. Binary objectness scores create a related problem because they do not distinguish positive boxes by localization quality. Geometric averaging is not a general solution either: nearby predictions should contribute only when their agreement is credible. Soft suppression and quality-aware scoring address parts of this process [14,15,18]. They do not jointly determine which sparse-radar proposals should remain available, how they should be ranked, and when their geometry can be refined safely.

We separate this problem into proposal availability, proposal quality, and proposal consistency. Availability determines whether weak evidence survives long enough to be examined. Quality concerns whether confidence reflects the localization of a positive box rather than object presence alone. Consistency describes whether neighbouring predictions and an independent detector support the same object hypothesis. These properties are connected but not interchangeable. Retaining more boxes does not improve their ordering, and averaging retained boxes can be harmful when local support is unreliable. A proposal-level method must therefore preserve evidence without treating every neighbour as equally trustworthy.

To study these decisions directly, we developed an expert-consistency-guided proposal-refinement framework that leaves the detector backbone unchanged. A candidate-preserving stream retains primary and complementary evidence that would otherwise be removed during early filtering. An independently trained radar expert then reweights the retained proposals according to cross-model agreement. Only supported primary proposals participate in restricted local voting, which limits the influence of unreliable neighbourhoods and leaves residual proposals unchanged. A separate localization-quality-aware branch aligns confidence with 3D intersection over union and provides an accuracy-oriented operating point. Both routes use the same PointPillars-style backbone, so their proposal behaviour can be compared without changing feature capacity.

Average precision alone does not show whether an improvement is repeated across datasets and training initializations. This distinction matters for radar because the evaluated datasets differ in return density, sensing geometry, coverage, and preprocessing. We therefore report two predefined endpoints. The strict route emphasizes positive paired differences across all dataset–seed comparisons, whereas the high-performance route emphasizes macro average precision. This separation makes the trade-off between consistent improvement and higher average accuracy explicit.

The main contributions of this study are as follows:

1. We formulate sparse-radar detection as a problem of proposal availability, localization-aware ranking, and geometric consistency. The resulting framework operates without changing the detector backbone.
2. We use expert consistency to reweight retained proposals and restrict local voting to boxes supported by independent radar evidence. A separate quality-aware route provides an operating point for higher average accuracy.
3. We evaluate both routes on four radar dataset conversions using three training seeds. The strict route improves all 12 matched comparisons, while the high-performance route raises macro AP from 35.3552 to 39.1214.

The remainder of the paper reviews related work, describes the two refinement routes, and reports the experimental protocol and results. The final sections discuss the observed mechanisms, practical boundaries, and computational cost.

2. Related Work

2.1. Radar-based 3D detection and representations

Automotive radar data can be encoded as spectra, range–angle maps, dense tensors, or post-detection point clouds. Each representation retains a different part of the measurement signal. Dense formats keep responses that may disappear during point extraction, at the cost of larger grids and specialized encoders. Post-detection point clouds are compact and compatible with established 3D detectors, but remain sparse and dependent on the sensor and preprocessing pipeline. Networks trained on radar spectra and dense tensors have shown that these measurements can support learned detection [21,22]. We use point-cloud conversions to provide the same proposal interface across all evaluated datasets. This also fixes the stage at which the methods are compared. Raw-signal encoding remains part of each dataset conversion, while refinement begins only after the detector has decoded its boxes.

Point-based, voxel-based, and pillar-based detectors handle irregular point sets in different ways. PointNet and PointNet++ operate directly on unordered points [1,2]. VoxelNet, SECOND, and submanifold sparse convolutions regularize the input while avoiding computation over most empty cells [3,4,20]. PointPillars encodes vertical pillars and scatters them into a bird's-eye-view (BEV) feature map [5]. Two-stage and center-based detectors add further geometric reasoning to improve proposal features [6,7,9]. Regardless of the backbone, final AP still depends on the ordering and removal of predicted boxes. We therefore retain a fixed PointPillars-style backbone and study the proposal decisions made after prediction.

2.2. Candidate preservation and expert-guided refinement

Two-stage detectors treat proposals as an explicit intermediate representation. Faster R-CNN generates regions before classification and box regression, while Cascade R-CNN refines localization over successive stages [10,17]. VoteNet applies learned Hough voting to aggregate point evidence around object centres [8]. These architectures differ from a single-stage radar detector, but share the idea that several uncertain observations can support a more stable object hypothesis.

Non-maximum suppression (NMS) turns a dense collection of hypotheses into a ranked detection list. Hard NMS removes overlapping boxes immediately. Soft-NMS instead reduces their scores [18], and learned NMS models duplicate removal as a proposal-reasoning problem [41]. Sparse R-CNN learns a fixed proposal set and updates its elements iteratively [19]. Suppression is especially consequential for sparse radar because different boxes may be supported by different subsets of the few returns from one object. Keeping only the highest-scoring box can discard useful localization evidence. Keeping every box creates duplicates and admits unsupported hypotheses. A bounded proposal set offers a practical compromise and leaves candidate evidence available for a later reliability check. These methods also intervene at different stages. Soft-NMS changes inference scores, learned NMS requires another trained module, and Sparse R-CNN changes the detector architecture. Our experiments instead apply refinement rules to a fixed detector output.

Box voting estimates consensus geometry from neighbouring predictions rather than selecting a single box. Weighted Boxes Fusion uses confidence-weighted coordinates for a related purpose [42]. Three-dimensional boxes require additional care because centres, dimensions, and periodic headings cannot be averaged in the same way. In our framework, preservation keeps candidate evidence available, while voting is restricted to proposals with independent expert support. The expert supplies only a support signal and does not add boxes to the final output. Training the expert independently also avoids deriving both a proposal score and its supporting evidence from the same prediction stream. M1 and M2 retain candidate evidence, M3 changes localization-aware ranking, and expert-guided voting is applied only after local agreement has been established.

2.3. Confidence and localization quality

Target assignment controls which locations learn objectness and box regression. FCOS and ATSS use explicit or adaptive assignment rules [12,13]. Focal Loss addresses class imbalance, while IoU-Net, Generalized Focal Loss, VarifocalNet, and TOOD incorporate localization quality or task alignment into the ranking score [11,14–16,40]. The distinction matters in sparse radar: a box may contain strong evidence of an object while its centre, heading, or dimensions remain inaccurate. Since NMS and thresholding follow the confidence order, this mismatch can affect AP. Small changes in a sparse-radar box can also produce large changes in 3D IoU. The quality target must therefore be computed from decoded positive boxes rather than assigned as a fixed class label. We study localization-aware scoring as a separate training branch, while retaining an objectness signal for assigned positives.

Quality-aware ranking and confidence calibration are related but distinct. Average precision depends mainly on ordering, while calibration compares the reported confidence with empirical correctness [45]. Detection calibration can additionally depend on location, scale, class, and localization uncertainty [43,44]. We report expected calibration error as a diagnostic for the accuracy-oriented route. The strict route is assessed through paired AP because its purpose is to test whether each matched evaluation improves, not whether confidence values are calibrated.

2.4. Radar benchmarks and multimodal context

K-Radar provides 4D radar measurements in varied weather, and Enhanced K-Radar examines density reduction and data accessibility [25,26]. View-of-Delft and TJ4DRadSet add urban road users and driving scenes [27,28]. V2X-Radar includes vehicle, infrastructure, and cooperative viewpoints [29]. These datasets differ in coverage, return density, sensing geometry, and labelling. Results obtained from one dataset or conversion may therefore reflect a particular measurement regime. We use the same detector family and matched training seeds across four conversions. The comparisons are kept paired within each dataset rather than combined into a shared leaderboard. Dataset-level AP values remain in their original evaluation contexts, while paired changes show whether an effect has the same direction across measurement regimes.

Multimodal detectors combine radar measurements with image features, spatial fusion, temporal alignment, or cross-modal objectives [30–35,38,39]. Image features can supply semantic information when radar observations are weak. The radar-only setting used here removes that source of information and makes proposal credibility depend on radar evidence alone.

Prior work has established strong backbones, adaptive assignment, localization-aware scores, suppression methods, and public radar benchmarks. Their interaction at the proposal level has received less attention under matched multi-seed and multi-dataset evaluation. We study that interaction with a fixed PointPillars-style backbone, the RDAR proposal stream as the reference, and separate routes for average accuracy and paired consistency.

3. Proposed Method

3.1. Problem formulation

Let a radar frame be represented by a point set $X = \{\mathbf{x}_n\}_{n=1}^{N_x}$, where each point contains spatial coordinates and the radar attributes supplied by the corresponding dataset conversion. The detector maps X to a set

$$\mathcal{P} = \{(\mathbf{b}_i, s_i, c_i)\}_{i=1}^{N_p}, \tag{1}$$

where $\mathbf{b}_i = (x_i, y_i, z_i, l_i, w_i, h_i, \theta_i) \in \mathbb{R}^7$ is a 3D oriented box, $s_i \in [0, 1]$ is its confidence, and c_i is its class. The experiments consider the car class and use the dataset-specific training and validation partitions frozen in the project protocol.

Conventional training minimizes classification, regression, and direction losses, whereas inference additionally applies score filtering and NMS. Our objective is not simply to increase the number of raw candidates. We seek a ranked set that satisfies three properties: plausible weak candidates remain available for recovery; confidence preferentially ranks better-localized boxes; and redundant hypotheses can provide geometric consensus rather than only suppressing one another. Reliability is then assessed by paired AP differences across datasets and seeds.

We distinguish *proposal availability*, *proposal quality*, and *proposal consistency*. Availability concerns whether a candidate reaches the final processing stages. Quality concerns whether its score agrees with its localization. Consistency concerns whether neighboring predictions support a common geometry. M1 and M2 primarily address availability, M3 addresses quality, and strict consistency refinement acts on the final proposal set. The division is functional: M1, M2, and the final gate/vote are proposal-level operations, while M3 changes training.

3.2. Baseline detector and framework overview

The common detector follows PointPillars [5]. Points are grouped into vertical pillars with a voxel size of $[0.25, 0.25, 5.0]$ m within the baseline range $[0, -25, -3, 60, 25, 2]$ m. At most 32 points are retained per pillar, with 16,000 training voxels and 40,000 test voxels. A PillarVFE maps each occupied pillar to 64 channels, PointPillarScatter places the features on the BEV grid, and a 2D backbone with block depths $[3, 5, 5]$, strides $[2, 2, 2]$, and channels $[64, 128, 256]$ produces the detection features. Three upsampling branches return 128 channels each and are concatenated before the anchor head.

The baseline head predicts classification, seven box residuals, and a two-bin direction label. Its loss weights are 1.0 for classification, 2.0 for localization, and 0.2 for direction. For the frozen Astyx configuration, the car anchor is approximately $[4.003, 1.800, 1.510]$ m with bottom height -0.180 m, and the positive/negative matching thresholds are 0.60/0.45. Dataset-specific configuration files provide the corresponding data paths and anchor statistics, while the network family and evaluation rule remain fixed.

The method first builds a shared proposal stream and then evaluates two endpoints. M1 retains weak but plausible primary proposals, and M2 adds a conservatively scored residual tail. Together, these two stages define the candidate-preserving RDAR baseline used throughout the experiments.

The strict route starts from the RDAR primary proposals. It changes their ranking through the expert-consistency gate and refines supported boxes by local voting. The high-performance route is separate: it trains the M3 quality-aligned confidence head and uses relaxed positive assignment with the accuracy-oriented prior. Voting is not appended to this configuration.

The shared proposal construction and the two operating routes are summarized in Fig. 1.

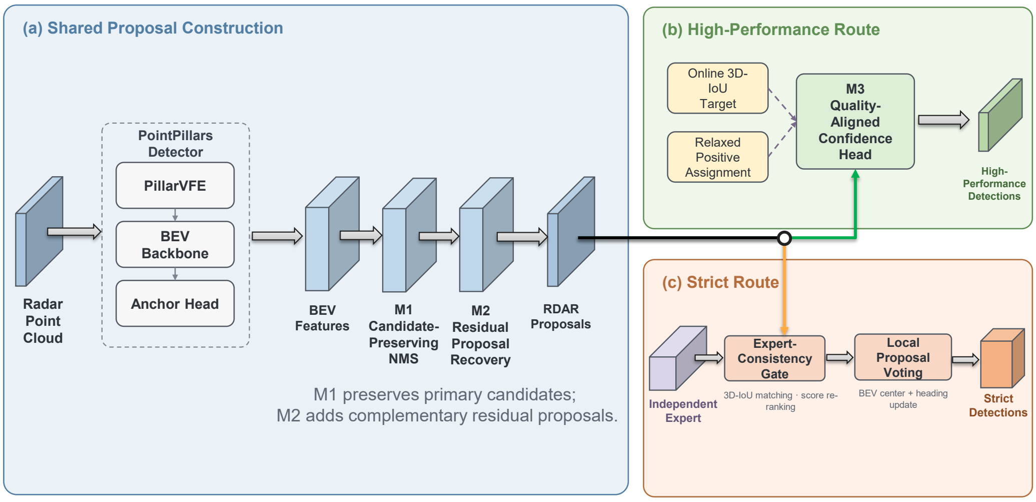

Figure 1. Overview of the proposal-refinement framework. PointPillars produces BEV features; M1 preserves candidate proposals and M2 adds complementary residual proposals to form the RDAR proposal stream. The high-performance route combines the M3 quality-aligned confidence head with an online 3D-IoU target and relaxed positive assignment. The strict route instead applies an independently trained expert for consistency gating, followed by local proposal voting.

The seed-2028 progressive analysis follows the construction of the high-performance route. By contrast, the formal strict comparison starts from RDAR-compatible proposals and selects for positive differences in all 12 dataset–seed pairs. The two analyses therefore measure different operating objectives.

3.3. Candidate preservation and recovery

3.3.1. M1: candidate-preserving NMS

The baseline configuration uses a score threshold of 0.10 and a very strict NMS IoU threshold of 0.01. Such a setting produces a compact output, but it can delete useful evidence before any complementary proposal is considered. M1 lowers the score threshold to zero and increases the NMS threshold to 0.50. The pre-NMS and post-NMS limits remain 4096 and 500, respectively. Thus M1 changes which boxes remain available, not the maximum reported proposal count. The design intent is to delay irreversible deletion until complementary evidence has been considered. M1 changes candidate availability rather than detector capacity or the final proposal budget.

Let π order boxes by score. Hard NMS accepts the highest remaining box and removes a box j whenever

$$\mathrm{IoU}_{3D}(\mathbf{b}_{\pi(i)}, \mathbf{b}_j) > \tau_{\mathrm{nms}}. \tag{2}$$

Increasing τ_{nms} from 0.01 to 0.50 delays this deletion and retains moderately overlapping alternatives. This resembles the preservation motivation of Soft-NMS [18], although our implementation does not use continuous score decay. The retained alternatives become inputs to recovery, quality-aware ranking, and voting.

Removing the initial score threshold does not mean every proposal is promoted to a high-confidence detection. The post-NMS limit remains finite, subsequent ranking uses the learned score, and residual additions in M2 are explicitly capped below the positive primary-score range. M1 is therefore a candidate- access operation: it creates the opportunity for later modules to use weak boxes without treating them as correct detections.

3.3.2. M2: residual proposal recovery

M1 cannot restore a hypothesis that exists only in a complementary prediction stream. M2 takes a primary set $\mathcal{P} = \{(\mathbf{b}_i, s_i)\}$ and a residual set $\mathcal{R} = \{(\mathbf{r}_j, t_j)\}$. Frame identifiers and ordering are validated before fusion. The residual boxes are sorted by t_j, and only the top $K_r = 50$ are processed.

The design intent is conservative recovery. A residual may contribute geometry when it supports a primary proposal, but its score must remain below the primary lane so that complementary evidence cannot override a confident primary prediction.

For each residual proposal, we find

$$i^\star(j) = \arg\max_i \mathrm{IoU}_{3D}(\mathbf{r}_j, \mathbf{b}_i). \tag{3}$$

When the best IoU is at least $\tau_r = 0.10$, the residual and primary geometries are combined using normalized score weights. For the six non-angular box parameters,

$$\widetilde{\mathbf{b}}_{j,1:6} = \frac{s_{i^\star}\mathbf{b}_{i^\star,1:6} + t_j\mathbf{r}_{j,1:6}}{s_{i^\star} + t_j}. \tag{4}$$

Heading is π-periodic for an oriented rectangular box. We therefore use doubled-angle circular averaging:

$$\widetilde{\theta}_j = \frac{1}{2}\,\mathrm{atan2}\left(\sum_{k\in\{i^\star,j\}} w_k \sin 2\theta_k, \sum_{k\in\{i^\star,j\}} w_k \cos 2\theta_k\right). \tag{5}$$

This avoids cancellation when equivalent headings fall on opposite sides of the numerical angle boundary. If the best overlap is below 0.10, the residual geometry is appended unchanged.

Recovered proposals are deliberately prevented from dominating the primary ranking. Let $s^+_{\min}$ be the smallest positive primary score in the evaluated prediction set and $t_{\max}$ the maximum residual score in the frame. The assigned recovery score is

$$s_j^{\text{rec}} = \left(\frac{s^+_{\min}}{2}\right)\left(0.5 + 0.5\frac{t_j}{\max(t_{\max}, \epsilon)}\right), \tag{6}$$

where adjacency denotes multiplication and $\epsilon = 10^{-12}$ prevents division by zero. The factor in parentheses lies in $[0.5, 1]$, so every residual score is at most half the smallest positive primary score.

3.4. *Localization-quality-aware confidence (M3 branch)*

M3 is an accuracy-oriented comparison branch, not a component of the primary expert-consistency endpoint. It is included because confidence ordering is a separate design axis from expert agreement; reporting it separately prevents the larger macro-AP gain from being mistaken for the main reliability claim.

Binary positive labels do not distinguish anchors by localization accuracy. M3 replaces the positive classification target with an online 3D-IoU target. For each positive anchor, the predicted residual and assigned regression target are decoded into $\mathbf{b}_i$ and $\mathbf{g}_i$. Their aligned quality is

$$q_i = \mathrm{IoU}_{3D}(\mathbf{b}_i, \mathbf{g}_i) \in [0,1]. \tag{7}$$

The IoU is detached from the classification graph, so the classification loss does not back-propagate through the IoU operation into box regression.

Directly using q_i can make a valid positive behave nearly like a negative while the regressor is immature. We retain a residual objectness component:

$$\widehat{q}_i = \rho + (1-\rho)q_i^{\gamma}, \qquad \rho = 0.55, \quad \gamma = 1.0. \tag{8}$$

Negatives receive zero and ignored anchors are excluded. The parameter ρ controls the minimum target for an assigned positive. At $\rho = 1$, the target reduces to binary objectness; at $\rho = 0$, it is pure IoU quality. The selected value retains both signals.

Let z_i be the classification logit and $p_i = \sigma(z_i)$. The quality focal term is

$$\mathcal{L}_{\mathrm{QF}} = \frac{1}{N_+} \sum_{i \in \mathcal{C}} |\widehat{q}_i - p_i|^{\beta}\, \mathrm{BCEWithLogits}(z_i, \widehat{q}_i), \qquad \beta = 2, \tag{9}$$

where the juxtaposition after the modulation factor denotes multiplication, $\mathcal{C}$ is the set of cared anchors, and N_+ is the number of positives, clipped below by one. This construction follows the general quality-aware principle of Generalized Focal Loss [14], but uses online aligned 3D IoU and an explicit objectness floor for sparse radar.

The full training loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{QF}} + 2\mathcal{L}_{\mathrm{box}} + 0.2\mathcal{L}_{\mathrm{dir}}. \tag{10}$$

M3 does not add a second inference branch. The original classification logit is trained to express the combined objectness–quality target, leaving the 4.830 M parameter count unchanged. The M3-only control retains the baseline assignment configuration and uses the candidate-preserving post-processing settings. The separate high-performance route combines M3 with 0.50/0.35 positive/negative assignment thresholds and the accuracy-oriented prior configuration.

The design intent is to make the ranking score reflect localization quality without discarding objectness. The role of M3 is ranking: a well-localized proposal should receive a larger target than a poorly localized proposal assigned to the same class. Better ranking can improve which proposal survives NMS and where a detection appears on the precision–recall curve. It does not guarantee that confidence is calibrated in every sense, so ECE is measured independently in Section 5.3.

3.5. Expert-guided refinement

The strict route applies two proposal-level operations to a primary RDAR set: an expert-consistency quality gate and local box voting. The expert is a stable BEV-gated detector with the same output parameterization. Its purpose is not to replace the primary detector but to provide a second assessment of whether a proposal is spatially supported. The strict route is selected by the paired no-regression gate described below, not by the performance of a single seed.

For reproducibility, the expert uses the same PointPillars-style radar input, class definition, box coder, and seven-parameter output head as the primary stream. Its configuration has a 64-channel PillarVFE, a 32-channel DensityAwarePillarGate in the pillar backbone, and a 64-channel BEV scatter. The BEV backbone uses three blocks with depths $[3, 5, 5]$, strides $[2, 2, 2]$, filters $[64, 128, 256]$, and three 128-channel upsampling paths. After concatenation, the expert applies a stable squeeze–excitation gate with reduction 8 and an initial bounded residual scale of 0.1, followed by the multi-scale BEV context module with hidden width 32, dilations $[1, 2, 4]$, and initial residual scale 0.1. The gate is channel-wise: for BEV feature x it computes $x \mapsto x(1 + r(2\sigma(g(\mathrm{GAP}(x))) - 1))$, where r is a learned bounded scalar initialized to 0.1. At the pillar level, the density-aware gate normalizes each pillar feature p_i, forms a scene context c_i by averaging normalized pillars from the same sample, and encodes the normalized point count $d_i = \mathrm{clip}(\log(1 + n_i)/\log(33), 0, 1)$. Specifically, $p_i = \mathrm{LayerNorm}(p_i^{\mathrm{raw}})$, and c_i is computed only from pillars whose first coordinate in the voxel-coordinate tensor equals the batch index of i; different batch samples therefore never share context. If a sample has no occupied pillars, the module returns it unchanged. It then computes

$$z_i = 2\,\sigma(W_2\ \mathrm{GELU}(W_1[p_i; c_i; d_i])) - 1, \qquad p_i' = p_i \odot (1 + r_p z_i), \tag{11}$$

where W_1, W_2 are biased linear layers with the framework's default parameter initialization, the hidden width is 32, and r_p is a learned scalar initialized to zero. The multi-scale BEV context first reduces the concatenated BEV tensor to 32 channels, applies depthwise 3×3 branches with dilations $1, 2, 4$, concatenates the branch outputs, projects them back to the original channel count using a bias-free 1×1 convolution followed by batch normalization, and adds the residual $r_c h(x)$ with scalar $r_c = 0.1$ at initialization. Each depthwise branch uses its own shared reduced input but does not share branch weights. These definitions make the expert modules explicit rather than relying only on YAML names. In the released implementation these correspond to the classes `DensityAwarePillarGate` and `MultiScaleBEVContext`; the stable channel gate is `StableSEGate2D`. Their source files are `density_aware_pillar_gate.py` and `base_bev_backbone.py` in the archived commit. The expert is trained independently with the same frozen split, augmentation policy, optimizer, 160-epoch schedule, and checkpoint epoch as the paired primary configuration; no additional labels or sensor inputs are used. The exact dataset-specific YAML files, checkpoint paths, and commit are included in the reproducibility archive. The fixed-expert checkpoint filenames, epoch, provenance paths, and SHA256 hashes are listed in `expert_checkpoint_manifest_20260730.md`. For the fixed-expert factorial audit, the expert checkpoint is held fixed within each conversion: seed 2027 is used for Astyx, TruckScenes, and V2X-Radar-V, and seed 2028 is used for K-Radar, while the primary prediction stream varies over seeds 2026, 2027, and 2028. This prevents the expert from implicitly tracking the seed of each primary cell.

3.5.1. Expert-consistency quality gate

Let $\mathcal{E} = \{(\mathbf{e}_j, u_j)\}$ be the expert predictions. The final $K_r = 50$ entries of the combined primary set are reserved for residual additions and are not gated. The gate is a support test rather than a second detector decision: agreement raises a proposal's rank, while disagreement lowers its rank without deleting the box. For every other primary proposal, the best expert match is

$$j^\star(i) = \arg\max_j \mathrm{IoU}_{3D}(\mathbf{b}_i, \mathbf{e}_j). \tag{12}$$

If the maximum IoU v_i is at least $\tau_e = 0.30$, the score becomes

$$s_i' = s_i^{1-\alpha} u_{j^\star}^{\alpha} v_i^{\eta}, \qquad \alpha = 0.30, \quad \eta = 0.25. \tag{13}$$

This geometric product requires agreement among primary confidence, expert confidence, and overlap. The small IoU exponent avoids allowing modest geometric noise to erase otherwise supported radar proposals. If $v_i < 0.30$, the proposal is retained but down-weighted:

$$s_i' = \kappa s_i, \qquad \kappa = 0.50. \tag{14}$$

Keeping unmatched proposals instead of deleting them preserves the conservative character of the pipeline. The gate affects ranking but does not move a box.

3.5.2. Local proposal voting

The voting stage refines geometry using the gated scores. Its design intent is to correct small center and heading errors while avoiding changes to box size, which could create artificial IoU gains. For each primary box i, a neighborhood is defined by

$$\mathcal{N}_i = \{j : \mathrm{IoU}_{3D}(\mathbf{b}_i, \mathbf{b}_j) \geq \tau_v\}, \qquad \tau_v = 0.24. \tag{15}$$

If $|\mathcal{N}_i| \leq 1$, the box is unchanged. Otherwise, $w_j = \max(s_j', 10^{-8})$ and

$$\mathbf{c}_{i,1:6} = \frac{\sum_{j \in \mathcal{N}_i} w_j \mathbf{b}_{j,1:6}}{\sum_{j \in \mathcal{N}_i} w_j} \tag{16}$$

defines the consensus geometry. Consensus heading is again calculated in doubled-angle space. The strict xy mode updates only the BEV center:

$$\mathbf{b}_{i,xy}' = (1-\lambda)\mathbf{b}_{i,xy} + \lambda \mathbf{c}_{i,xy}, \qquad \lambda = 0.40. \tag{17}$$

Heading is interpolated circularly with weights $1-\lambda$ and λ. Height, center z, length, width, and height remain unchanged. Restricting the update prevents the consensus from artificially shrinking or expanding boxes to satisfy the evaluation threshold.

The IoU neighborhood threshold is lower than the 0.50 evaluation threshold. This is intentional: multiple radar hypotheses for the same vehicle can have modest 3D overlap because of height and dimension noise while still agreeing in BEV. Threshold sensitivity is nevertheless evaluated over a closed grid rather than justified only after observing the selected setting. The final 50 residual proposals remain outside the voting pool, ensuring that consistency refinement cannot undo M2 by moving or suppressing the recovered tail.

The stages preserve a fixed division of responsibility. M1 determines which primary candidates survive, and M2 adds complementary residual proposals. In the strict route, expert consistency changes only the ranking of primary proposals. Local voting then updates only the BEV center and heading of supported primary boxes. Residual proposals never enter the gate or voting pool, and voting does not change box dimensions.

3.6. Operating routes and selection criteria

The *RDAR baseline* denotes M1 followed by M2 under the frozen recovery protocol. The *M3-only control* applies M3 and provides the intermediate trainable comparison. The *high-performance route* uses the same quality-focal formulation with relaxed positive assignment and the accuracy-oriented prior configuration. It is chosen by macro AP across the four datasets. Its intended use is to show the performance available when average accuracy is prioritized.

The *strict route* applies the expert gate and local vote. Let $A^m_{d,r}$ be AP for method m, dataset d, and seed r. A candidate is strict only when

$$\Delta_{d,r} = A^m_{d,r} - A^{\text{RDAR}}_{d,r} > 0 \quad \forall (d,r) \in \mathcal{D} \times \mathcal{S}, \tag{18}$$

where $|\mathcal{D}| = 4$ and $|\mathcal{S}| = 3$. This definition contains no tolerance for a small negative difference. It is stronger than requiring a positive macro mean, but remains bounded to the tested data, splits, and seeds.

Selection by Eq. (18) is not a claim that the strict route has the highest AP. Indeed, the high-performance route has a larger macro mean. The two routes answer different engineering questions: whether a configuration offers the largest average gain and whether it avoids a regression in the available paired tests.

Table 1 lists the exact operations enabled by every reported configuration. It should be consulted when comparing the M3-only control, high-performance route, and strict route because they optimize different objectives and are not successive aliases for one model. In particular, relaxed positive assignment belongs to the high-performance route, whereas expert gating and local voting define the strict route.

Table 1. Definitions of the operating routes and controls. M1 and M2 form the shared candidate-preserving proposal stream; the remaining columns identify the additional scoring or refinement operations.

Configuration	M1+M2 stream	M3 score	Expert gate	Local vote	Selection objective
RDAR baseline	Yes	Binary objectness	No	No	Reference
M3-only control	Yes	IoU-aware quality	No	No	Isolate quality ranking
High-performance route	Yes	IoU-aware quality + relaxed assignment	No	No	Maximize macro AP
Strict route	Yes	RDAR score	Yes	Yes	Positive paired differences

3.7. Algorithmic summary

For clarity, the complete procedure is summarized below.

Step 1: Encode the radar frame with PillarVFE, scatter pillar features to BEV, and predict anchor classification, box residuals, and direction.

Step 2: During M3 training, decode positive boxes, compute aligned 3D IoU, form the soft target in Eq. (8), and optimize the combined loss. The baseline route retains binary classification.

Step 3: Apply candidate-preserving NMS with zero initial score threshold, $\tau_{\text{nms}} = 0.50$, and a 500-box post-NMS limit.

Step 4: Retain the top 50 complementary residual proposals. Fuse a residual box with its best primary match when 3D IoU is at least 0.10; otherwise append it. Assign the conservative score in Eq. (6).

Step 5: For the strict route, match each non-residual primary proposal to the expert stream, apply Eq. (13), and down-weight unmatched boxes by 0.50.

Step 6: Form local neighborhoods at $\tau_v = 0.24$ and update BEV center and heading using Eq. (17). Return the residual tail unchanged.

Step 7: Rank the resulting predictions and evaluate them with the identical AP_{R40} implementation used for every compared method.

3.8. Training and implementation details

Training uses the OpenPCDet codebase and an Adam one-cycle optimizer for 160 epochs. The frozen baseline configuration uses a per-GPU batch size of four, initial learning rate 0.003, weight decay 0.01, momentum 0.9, one-cycle momenta $[0.95, 0.85]$, warm-up fraction 0.4, division factor 10, and gradient norm clipping at 10. Random world flips, rotations in $[-\pi/4, \pi/4]$, and scaling in $[0.95, 1.05]$ are applied during training. Ground-truth sampling is disabled in the frozen radar configuration used for the controlled comparisons.

All models are trained independently with seeds 2026, 2027, and 2028. The formal comparison uses the same data partitions, class definition, metric, checkpoint epoch, and evaluation code for each paired method. Proposal-level scripts validate frame order before combining prediction files. This check is important because an apparently valid fusion between misaligned frame lists could silently produce meaningless boxes.

The progressive construction study is intentionally narrower. It holds the seed at 2028 and adds M1, M2, quality alignment, and the high-performance calibration variant sequentially. It is used to identify the contribution scale of the stages, not to establish that every intermediate module is positive for all seeds. The formal strict result instead evaluates the final strict route on all 12 paired cells.

Computational cost is reported in three parts. Trainable parameters distinguish model capacity. A fixed profiler measures detector forward execution including OpenPCDet post-processing but excluding AP computation and result serialization. A separate timer measures only the local voting loop and excludes prediction-file input/output. Keeping these measurements separate prevents a fast detector result from hiding an expensive post-processing stage.

3.9. Implementation safeguards

The implementation uses four safeguards that define the scope of the claim. Prediction files must have identical frame order; residual scores are bounded by Eq. (6) and kept in the final 50-entry tail, outside the gate and voting pool; and residual fusion and voting use doubled-angle averaging for the π-periodic heading. The strict vote updates only x, y, and heading, while z, l, w, and h remain unchanged. Finally, one global threshold setting is used across all four datasets; dataset-specific compensation screens are reported only as diagnostics. These constraints keep the comparison auditable and prevent the reliability endpoint from becoming a collection of independently tuned dataset-specific procedures.

4. Experimental Protocol

4.1. Research questions

We ask five questions. RQ1 tests whether the strict route improves RDAR in every matched dataset–seed comparison. RQ2 isolates the contribution of each component, and RQ3 examines sensitivity to expert and threshold selection. RQ4 compares AP with calibration and controlled point loss. RQ5 reports the computational cost of both routes.

4.2. Datasets and protocol

We evaluate radar-only car detection on four datasets. Astyx provides an automotive radar point-cloud benchmark [23]. MAN TruckScenes covers autonomous trucking scenes and a sensor geometry distinct from passenger-car platforms [24]. For V2X-Radar-V, we use the vehicle-side subset of a cooperative 4D radar collection. K-Radar contains 4D radar measurements acquired in varied weather [25,29]. The experiment records refer to the frozen conversions as Astyx, TruckScenes-mini, V2X-Radar-V-400, and K-Radar-400. We use shorter names in the text. The experiment package retains the exact conversion names, configuration files, and frozen outputs.

The frozen splits contain 300/100 train/validation frames for Astyx and 320/80 for each remaining conversion, with 531, 3950, 268, and 198 validation car instances, respectively. The records provide frame-level point-cloud, calibration, image, and annotation fields but no scene or sequence identifiers. We found no exact train–validation overlap in the available point-cloud IDs. We also applied a conservative adjacency check: a symmetric ± 1 difference in `point_cloud.pc_idx`. It flagged 82/100 Astyx, 4/80 TruckScenes, 1/80 V2X-Radar-V, and 80/80 K-Radar validation frames. The converted records do not contain ordering metadata, so this check cannot establish temporal adjacency. It also cannot rule out correlations from the same scene or vehicle. Excluding all flagged frames would leave 18, 76, 79, and 0 validation frames, making K-Radar AP impossible to estimate. Our evaluation is therefore limited to the frozen frame-level protocol. We do not claim scene-disjoint, vehicle-disjoint, cross-initialization, or deployment generalization. The "mini" and "400" suffixes name the conversion protocol, not new dataset releases.

The primary metric is car AP_{R40} at 0.50 3D IoU, evaluated with a common implementation and class mapping. AP is reported on a 0–100 scale, so a difference of 1.0 is one AP point. Because AP is computed over the complete validation set and is not additive over frames, frame and instance counts describe evaluation scope rather than define per-frame confidence intervals. The paired and cluster-resampling summaries describe only these finite frozen sets.

The formal experiment uses four datasets and three seeds (2026, 2027, and 2028). Each model is trained independently under the same schedule and evaluated with the same class mapping and code. We report the mean and sample standard deviation across seeds. For the strict route, comparisons are paired within each seed. The primary detector and expert are trained independently, but use the same data protocol and seed. This pairing makes individual regressions visible without introducing a cross-seed difference. We examine that issue separately by holding the expert fixed across seeds. This diagnostic tests sensitivity, not initialization-independent robustness.

The main claims rely on the formal three-seed comparison. The seed-2028 progressive study is used only to examine the construction path. Calibration, point dropout, threshold transfer, runtime, and qualitative cases probe model behavior and practical limits.

Table 2 summarizes the conversions, inputs, spatial ranges, and training seeds. Acquisition settings differ, but the class mapping, AP_{R40} metric, 0.50 3D-IoU threshold, and paired-seed rule are shared.

Table 2. Dataset scope and common protocol for the formal four-dataset, three-seed evaluation.

Dataset	Frozen conversion	Platform or acquisition setting	Model input	Spatial range $[x, y, z]$ (m)	Formal seeds
Astyx	Astyx	Automotive passenger-vehicle scenes	Radar point cloud	$[0, 60] \times [-25, 25] \times [-3, 2]$	2026, 2027, 2028
MAN TruckScenes	TruckScenes-mini	Autonomous trucking scenes	Radar point cloud	$[0, 60] \times [-25, 25] \times [-3, 2]$	2026, 2027, 2028
V2X-Radar-V	V2X-Radar-V-400	Vehicle-side cooperative-perception subset	4D radar point cloud	$[0, 60] \times [-25, 25] \times [-3, 2]$	2026, 2027, 2028
K-Radar	K-Radar-400	Driving scenes under varied weather conditions	4D radar point cloud	$[0, 60] \times [-25, 25] \times [-3, 2]$	2026, 2027, 2028

All rows use the paper's frozen car-class mapping and are evaluated with AP_{R40} at a 3D-IoU match threshold of 0.50. Each method is trained independently for the three listed seeds, and paired method–RDAR differences are calculated within the same dataset and seed. The spatial range is the common model input range recorded in the frozen configurations, not a claim about the physical maximum range of each sensor.

4.3. Baselines and comparison routes

The conventional PointPillars-style detector is the architectural reference. RDAR, comprising M1 and M2, is the paired baseline for the formal comparison. The M3-only control isolates M3, and the high-performance route combines M3 with relaxed assignment and the accuracy-oriented prior. The strict route adds expert-consistency score refinement and conditional geometry voting to RDAR-compatible primary proposals. Frozen-output controls include standard box voting, Gaussian Soft-NMS, and weighted box fusion. A gate–vote factorial comparison and a fixed-expert audit are used to attribute the strict-route effect.

4.4. Metrics and statistical summaries

For a ranked list at a fixed 3D-IoU match threshold, precision and recall after the first k predictions are

$$\mathrm{Prec}(k) = \frac{\mathrm{TP}(k)}{\mathrm{TP}(k) + \mathrm{FP}(k)}, \qquad \mathrm{Rec}(k) = \frac{\mathrm{TP}(k)}{\mathrm{TP}(k) + \mathrm{FN}(k)}. \tag{19}$$

AP_{R40} samples the interpolated precision envelope at 40 recall positions. Because confidence determines the traversal order, M3 can change AP even when the set of decoded boxes is similar. Because voting changes geometry, the strict route can also change which predictions cross the 0.50 IoU matching threshold.

The macro result for method m is the unweighted average of the four dataset-level seed means:

$$\overline{A}^{m}_{\mathrm{macro}} = \frac{1}{4} \sum_{d \in \mathcal{D}} \left(\frac{1}{3} \sum_{r \in \mathcal{S}} A^{m}_{d,r} \right). \tag{20}$$

This choice gives each dataset equal influence regardless of the number of frames or objects. It avoids allowing the largest validation partition to dominate, but it does not imply equal operational importance.

Standard deviations describe variation across the three seeds; they are not confidence intervals for dataset performance. For the strict comparison, we report all 12 paired differences, their mean and median, a bootstrap interval, and an exact sign test. We place greater weight on the effect sizes and the smallest paired gain. Threshold selection and evaluation use the same four-dataset protocol, so the result describes consistency within this protocol rather than held-out performance. Resampling the four dataset means gives a descriptive interval of [0.5464, 1.4834] AP.

5. Results

5.1. Main comparison

Table 3 compares the two operating endpoints with RDAR and the M3-only control over three seeds. The high-performance route is evaluated by macro AP and calibration, whereas the strict route is evaluated by its paired AP difference from RDAR in each dataset–seed cell.

Table 3. Main three-seed comparison of RDAR, the M3-only control, the high-performance route, and the strict route.

Route	Astyx	TruckScenes	V2X-Radar-V	K-Radar	Macro	ΔMacro	Positive paired cells
RDAR	32.8347 ± 1.4689	16.3671 ± 1.7480	41.7029 ± 1.1490	50.5163 ± 2.0546	35.3552	–	–
M3-only	35.8438 ± 1.0680	16.9661 ± 1.2785	**44.4122 ± 2.5011**	57.6910 ± 2.5304	38.7283	+3.3731	11/12
High-performance	**36.9801 ± 0.9748**	16.8063 ± 1.1764	44.3786 ± 0.5122	**58.3208 ± 1.6943**	**39.1214**	**+3.7662**	11/12
Strict	33.6898 ± 1.7215	**17.2878 ± 1.7127**	42.1245 ± 1.0852	52.2091 ± 2.5654	36.3278	+0.9726	**12/12**

Values are AP_{R40} mean ± sample standard deviation over seeds 2026, 2027, and 2028 at a 3D-IoU threshold of 0.50. Macro is the unweighted mean of the four dataset means, and ΔMacro is measured from RDAR. Bold values mark the largest result in each dataset or aggregate column. "Positive paired cells" counts the dataset–seed cells with an AP increase relative to RDAR. The M3-only control and high-performance route correspond to the frozen quality-aware and accuracy-oriented configurations described above.

The strict route raises macro AP from 35.3552 to 36.3278 (+0.9726). It also improves all 12 matched dataset–seed cells. The high-performance route reaches 39.1214 macro AP (+3.7662), with its largest gain on K-Radar. Its only decrease is on TruckScenes at seed 2027, from 18.3845 to 17.1075 AP. The M3-only control reaches 38.7283 macro AP and has the same negative cell. The paired differences are plotted in Fig. 2 and listed in Table 4.

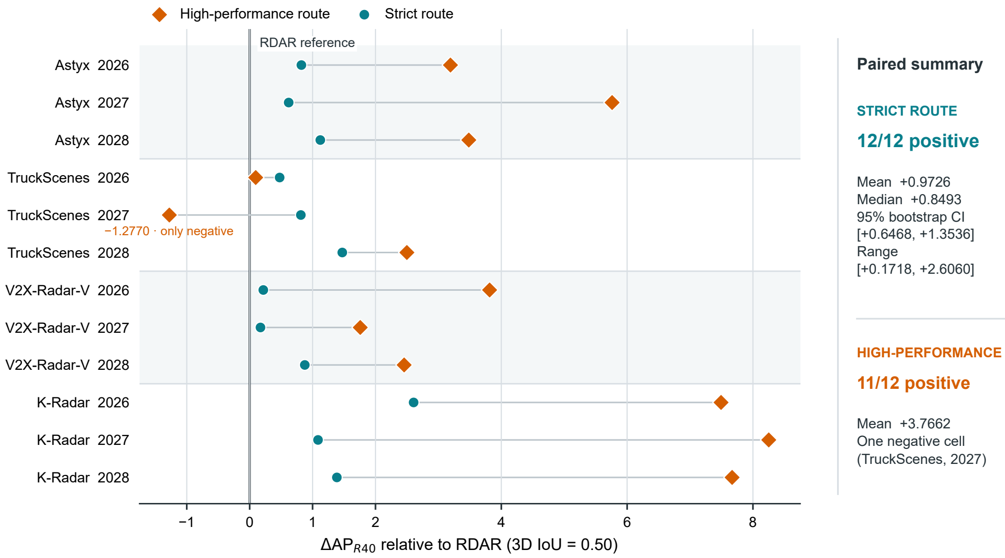

Figure 2. Paired AP differences relative to RDAR across the four datasets and three seeds. The strict route remains positive in every tested dataset–seed cell, whereas the high-performance route contains one negative cell.

Table 4 gives the complete matched-seed audit for the strict route.

Table 4. Complete matched-seed reliability audit for the strict route.

Dataset	Seed	RDAR AP	Strict AP	ΔAP
Astyx	2026	32.7281	33.5503	+0.8222
Astyx	2027	31.4220	32.0422	+0.6202
Astyx	2028	34.3540	35.4768	+1.1228
TruckScenes	2026	15.4127	15.8892	+0.4765
TruckScenes	2027	18.3845	19.1980	+0.8135
TruckScenes	2028	15.3041	16.7762	+1.4721
V2X-Radar-V	2026	40.7802	40.9969	+0.2167
V2X-Radar-V	2027	42.9899	43.1617	**+0.1718**
V2X-Radar-V	2028	41.3385	42.2148	+0.8763
K-Radar	2026	51.3450	53.9510	**+2.6060**
K-Radar	2027	48.1767	49.2631	+1.0864
K-Radar	2028	52.0271	53.4132	+1.3861

Strict AP is the RDAR value plus the frozen matched difference for the strict route. Across the 12 cells, the mean, median, minimum, and maximum ΔAP are 0.9726, 0.8493, 0.1718, and 2.6060. The nonparametric bootstrap 95% interval for the mean is [0.6468, 1.3536]. The exact sign test gives one-sided $p = 0.000244$ and two-sided $p = 0.000488$; these tests are auxiliary summaries for the finite evaluated cells.

All paired differences are positive. Removing the largest gain does not change this result. The smallest margin is 0.1718 AP on V2X-Radar-V at seed 2027. The table note and experimental protocol give the remaining effect-size and uncertainty summaries.

5.2. Component and Control Analysis

We applied three fixed-parameter controls to the same frozen RDAR predictions. They use the same AP code and require neither expert predictions nor additional labels. In Table 5, standard box voting is positive for all three seeds on Astyx and V2X-Radar-V. Gaussian Soft-NMS and weighted box fusion do not meet this criterion on any dataset. None matches the strict route across all four datasets.

Table 5. Three-seed controls for generic proposal post-processing. Values are mean AP differences relative to RDAR.

Control	Astyx	TruckScenes	V2X-Radar-V	K-Radar	Datasets positive in all three seeds
Standard box voting	+0.4723	-0.0831	+0.5342	+0.0097	2/4
Gaussian Soft-NMS ($\sigma = 0.5$)	-0.1831	+0.0909	-0.2098	+0.0196	0/4
Weighted box fusion (IoU = 0.5)	-0.0118	-0.1305	-0.2474	-0.1896	0/4
Strict route	+0.8551	+0.9207	+0.4216	+1.6928	4/4

All methods start from the same frozen RDAR prediction files and evaluator; control parameters were fixed before three-seed aggregation.

Across the 12 primary cells, vote-only changes mean AP by -0.0274 and improves 6/12 cells. We then fixed one expert checkpoint within each dataset instead of matching the seed. Under this test, gate-only improves 11/12 cells, and gate-plus-vote improves 10/12. Both regressions occur on Astyx. The matched strict result is narrowest on V2X-Radar-V: its mean gain is 0.4216 AP, with two seed-level gains below 0.22 AP.

5.2.1. Progressive construction

Fig. 3 follows the seed-2028 construction path on the four datasets. M1 has its clearest effect on Astyx, whereas M2 changes the result only slightly. M3 produces the largest macro step, especially on K-Radar. This analysis uses one seed.

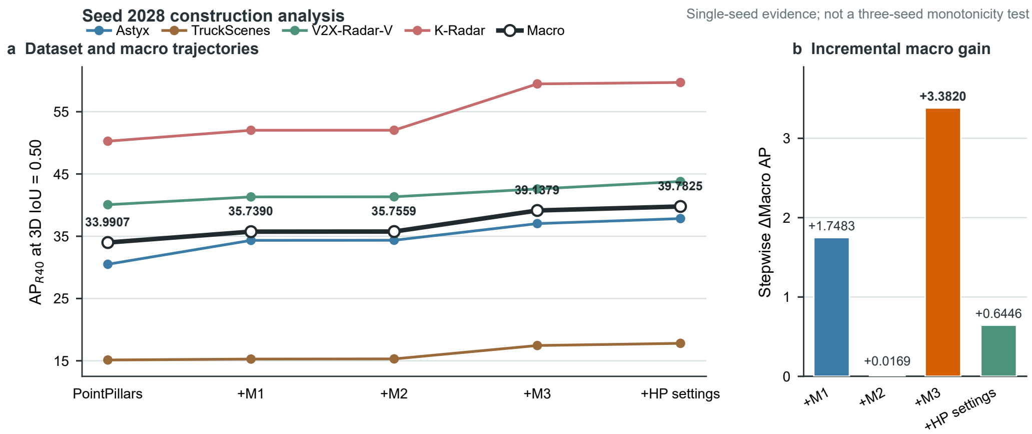

Figure 3. Progressive construction on seed 2028. M1 and M2 preserve and recover candidate evidence, while M3 supplies the dominant accuracy-oriented gain; the plot is a mechanism analysis for one seed rather than a multi-seed monotonicity claim.

Relative to the preceding configuration, M1 adds 1.7483 macro AP and M2 adds 0.0169. M3 raises macro AP from 35.7559 to 39.1379. The corresponding gains are 2.6778 AP on Astyx, 2.1489 on TruckScenes, 1.2594 on V2X-Radar-V, and 7.4416 on K-Radar. The final accuracy-oriented adjustment adds 0.6446 AP.

5.3. Robustness and Threshold Sensitivity

Fig. 4 summarizes calibration, controlled point loss, voting thresholds, and the number of positive paired cells.

ECE compares mean confidence with empirical correctness within confidence bins [45]. The high-performance route lowers ECE in all 12 three-seed comparisons, with a mean reduction of 0.0504. At seed 2028, the reductions are 0.0749 on Astyx, 0.0407 on TruckScenes, 0.0354 on V2X-Radar-V, and 0.0845 on K-Radar. M3-only gives the lowest ECE on V2X-Radar-V; the high-performance route is lowest on the other datasets. In contrast, the strict route increases ECE in all 12 comparisons by 0.0270 on average.

The range-bin audit gives the strict route 26 wins, 13 ties, and 9 losses. Its sparsity audit gives 18 wins, 19 ties, 8 losses, and 3 missing cells. The high-performance route records 17

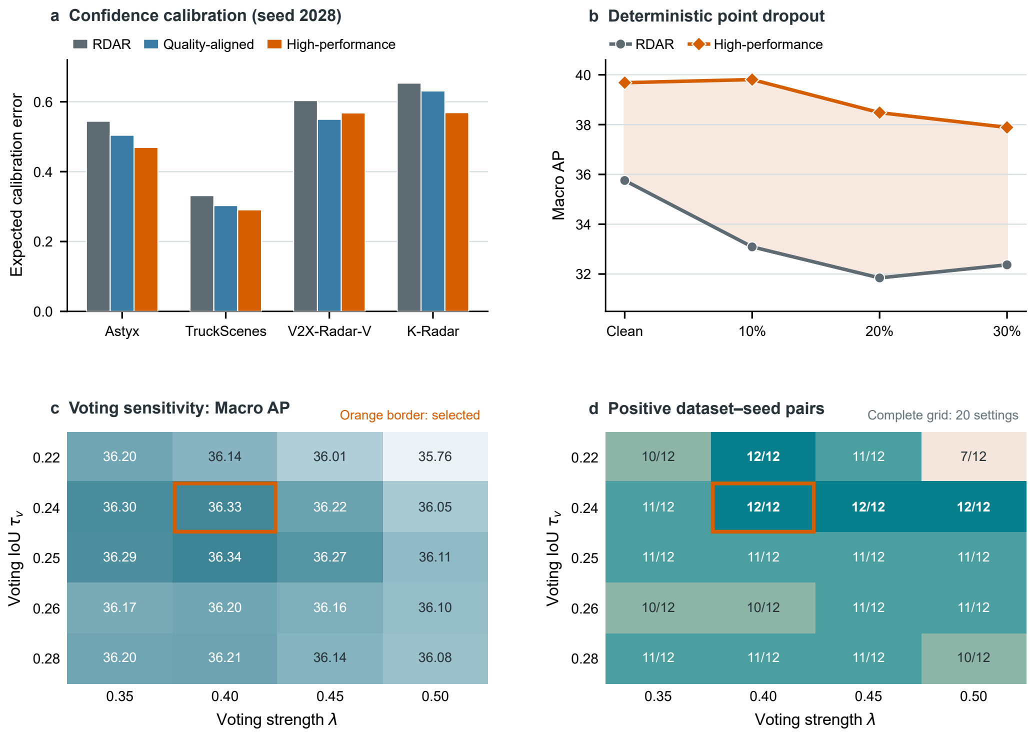

Figure 4. Diagnostic and boundary analyses. Panels compare calibration, controlled point loss, voting thresholds, and the number of positive paired cells across the reported routes.

range-bin wins, 10 ties, and 21 losses. The bin-level counts include both improvements and regressions.

5.3.1. Point-dropout stress test

We dropped radar points before voxelization with a fixed seed of 3407. No model was retrained or adapted. The high-performance route remains ahead at 10%, 20%, and 30% dropout. Its changes from the clean checkpoint are $+0.1242$, -1.2030, and -1.7970 AP. The corresponding RDAR changes are -2.6651, -3.9114, and -3.3868, a mean difference in degradation of 2.3624 AP.

Across datasets and dropout rates, the high-performance route exceeds RDAR in $11/12$ cells. The exception is TruckScenes at 30%: 11.9555 versus 12.0161 AP, a difference of -0.0606. Adding an exploratory vote fixes this cell but reduces the original result in $8/12$ cells, so we leave the endpoint unchanged.

5.3.2. Voting-threshold sensitivity

Fig. 4(c,d) shows the complete grid over voting IoU and strength. Three neighboring settings also remain positive in all 12 dataset–seed pairs: $(0.22, 0.40)$ reaches 36.1364 macro AP, $(0.24, 0.45)$ reaches 36.2165, and $(0.24, 0.50)$ reaches 36.0479. The highest grid value, 36.3393 at $(0.25, 0.40)$, is positive in 11 of 12 pairs. The selected $(0.24, 0.40)$ setting reaches 36.3278 macro AP and remains the best setting under the 12-of-12 criterion.

5.3.3. Held-out threshold transfer

For leave-one-dataset-out transfer, the conservative screen keeps settings with nine positive training differences and then selects the highest mean. The unconstrained screen selects by training mean alone. Without further tuning, the conservative rule gives positive held-out means for all four datasets and $11/12$ positive seed differences. The unconstrained rule yields one negative cell: −0.2945 AP on V2X-Radar-V at seed 2027. This audit is separate from the formal threshold-selected result.

5.4. Qualitative Cases

Fig. 5 compares RDAR and strict-route boxes for three selected borderline misses. The examples show how center and heading updates can cross the 0.50 matching threshold without changing box dimensions. They do not indicate how often this occurs.

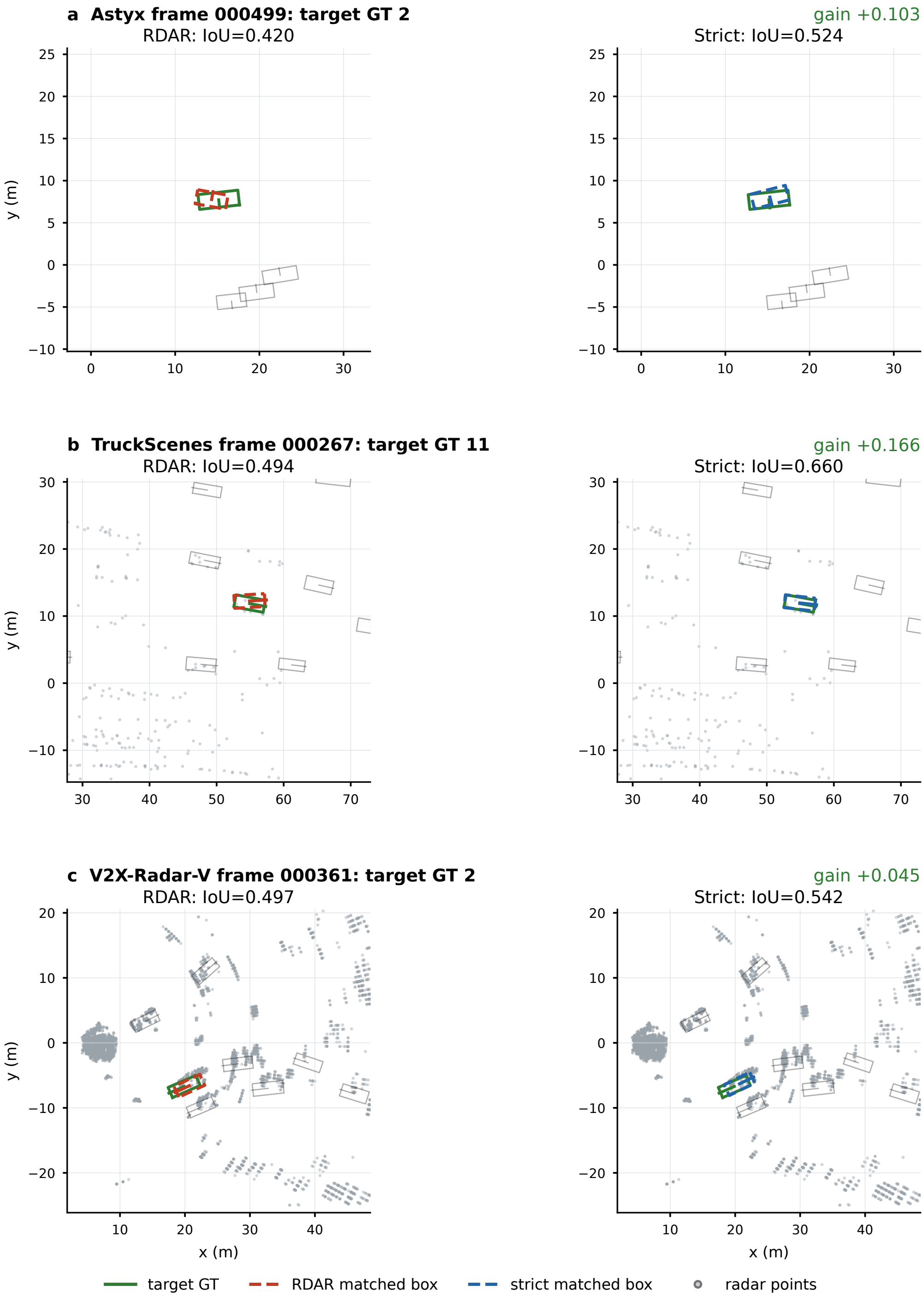

Figure 5. Qualitative BEV cases comparing RDAR and the strict route against the same ground-truth targets. The constrained update changes center and heading while keeping box dimensions fixed.

Conditional voting raises IoU from 0.4203 to 0.5235 in Astyx frame 000499. It raises IoU from 0.4940 to 0.6599 in TruckScenes frame 000267 and from 0.4970 to 0.5420 in V2X-Radar-V frame 000361. In each case, the restricted xy and heading update turns a borderline localization into a valid match without resizing the box.

5.5. Computational Cost

Table 6 reports parameter counts and timing measurements. M3 changes the training target rather than network width, so the M3-only and high-performance detectors retain the RDAR parameter count. Voting adds no trainable parameters, but it adds proposal-refinement time. We report detector forward time, isolated voting time, and endpoint time separately.

Table 6. Component-level parameter and runtime profile on an NVIDIA GeForce RTX 3090.

Configuration	Parameters	Parameter interpretation
RDAR detector	4.830 M	Primary detector and frozen internal baseline
Quality-aligned detector	4.830 M	M3 changes supervision, not network width
High-performance detector	4.830 M	Separate trained accuracy-oriented endpoint
Stable expert detector	4.868 M	Auxiliary detector used by the strict route
Strict voting operation	+0	Proposal-level operation with no trainable weights

Profiled stage	Astyx	TruckScenes	V2X-Radar-V	K-Radar
RDAR detector forward	4.740 ± 0.297	3.851 ± 0.311	4.834 ± 1.092	5.887 ± 2.409
High-performance detector forward	8.263 ± 1.163	7.880 ± 1.257	8.124 ± 1.334	8.601 ± 1.471
Strict voting loop	22.7095 ± 0.1031	22.0086 ± 0.1826	22.5393 ± 0.0471	22.8485 ± 0.1735

Hardware: NVIDIA GeForce RTX 3090. Detector-forward profiling uses batch size 4 and excludes AP evaluation, result serialization, expert inference, and prediction-file I/O. Independently profiled rows must not be summed as an end-to-end strict-route latency.

Both M3-based detectors have 4.830 M parameters, the same as RDAR. The expert used by the strict route has 4.868 M parameters, and voting has no trainable weights. On the RTX 3090, the high-performance detector remains below 9 ms per frame but is slower than RDAR. The isolated 500-proposal voting loop takes about 22–23 ms. It constructs 3D IoUs on the GPU, then visits the local neighborhoods in a Python loop.

For completeness, an exploratory batch-1 profile runs primary forward, expert forward, and post-processing in one process on an RTX 4080 SUPER. Mean times are 16.08 ms on Astyx, 23.11 ms on TruckScenes, and 21.62 ms on V2X-Radar-V. These values correspond to 62.2, 43.3, and 46.3 Hz, with 197–201 MB peak allocated memory. The sampled Astyx frames return 20–27 proposals, unlike the 500-proposal RTX 3090 component profile, so the two sets of timings are not combined. In a separate scaling test, voting takes 1.5–1.6 ms for 50 proposals and 9.7–10.1 ms for 500. Peak incremental allocation is approximately 6.7 MB.

6. Discussion

No single route dominates across all criteria. The high-performance route has the larger mean AP gain and better calibration, but one matched comparison is negative. The strict route improves all 12 dataset–seed cells, with a smaller mean gain and poorer calibration. In practice, the choice is between maximizing average detection quality and avoiding a regression under the present protocol.

6.1. Mechanism and Relation to Prior Work

Candidate preservation leaves alternative hypotheses available after the first suppression step, while residual recovery restores proposals that would otherwise disappear. In the seed-2028 construction, most of the macro-AP increase comes from M3. Localization-aware ranking is therefore the main contributor in this run. Since the progressive experiment uses only one seed, the same ordering cannot be assumed across runs.

The controls help explain why the strict route behaves differently from generic post-processing. Vote-only improves 6 of the 12 primary cells. Neither fixed Soft-NMS nor weighted box fusion produces three positive seeds on any dataset. When one expert checkpoint is fixed within each dataset, gate-only is positive in 11/12 cells and gate-plus-vote in 10/12. These results place most of the stability benefit in the independent expert check, with voting playing a more selective role. They do not isolate the gate completely because the fixed-expert and matched-expert protocols differ.

The expert check also separates the strict route from existing post-processing methods. Soft-NMS adjusts scores from proposal overlap, whereas weighted box fusion averages coordinates using detector confidence [18,42]. Here, an independently trained expert determines whether a primary proposal can be re-ranked or updated. M3 acts earlier by aligning the ranking score with localization quality, an idea shared with IoU-aware confidence methods [14,16]. Our experiments examine how these two operations interact on a fixed PointPillars-style backbone; they do not show that either class of method can be replaced.

6.2. Operational Trade-Offs and Boundaries

AP consistency and calibration move in opposite directions. The high-performance route lowers ECE in all 12 comparisons, whereas the strict route raises it in all 12. Strict-route scores should therefore be recalibrated before they are used as probabilities for risk-sensitive thresholding. The high-performance route is the more natural choice when ranking and confidence quality matter together.

Robustness is similarly uneven. The range and sparsity audits contain both wins and losses, and TruckScenes regresses at 30% point dropout. Point density, return geometry, or the localization errors presented to the expert may account for some of this variation, but the current audits do not separate these factors. The gains are consequently specific to the tested datasets and operating regimes.

The computational profiles add another practical constraint. The high-performance detector retains the RDAR parameter count but has higher forward latency. The strict route also requires expert inference, and its 500-proposal Python voting loop takes about 22–23 ms on the RTX 3090. Because the exploratory RTX 4080 SUPER profile uses different proposal counts, it is not directly comparable with the component-level RTX 3090 measurements. A deployment decision must account for calibration, the cost of a regression, and the available latency budget.

6.3. Limitations and future work

The conclusions are bounded by radar-only car detection, four dataset conversions, three seeds, and one PointPillars-style detector. The smallest strict gain is 0.1718 AP, and the voting threshold is selected within the same evaluation protocol. Leave-one-dataset-out selection reduces this dependence but still produces one negative held-out seed. The frame manifests do not allow a scene-disjoint audit, and the progressive construction uses only one seed. The qualitative examples are selected successful corrections, not an estimate of how often such cases occur.

Another gap is the lack of a protocol-matched comparison with recent public radar-only detectors. Published scores based on different dataset conversions, spatial ranges, and evaluators cannot be inserted fairly into the present table. The most useful next tests are development-set threshold selection, scene-aware splits, additional detector families and classes, and reproduced public baselines under the same evaluator. Post-hoc calibration of the strict route and a batched or sparse-neighbourhood voting implementation would address the two main deployment costs identified here.

7. Conclusion

We introduced a proposal-level refinement framework for radar-only 3D car detection that leaves the detector backbone unchanged. It retains alternative candidates, uses localization quality in proposal ranking, and checks proposed updates against an independently trained expert. Over four dataset conversions and three seeds, the strict route improved AP in all 12 matched dataset–seed comparisons, with an average gain of 0.9726 AP. The high-performance route achieved the larger macro-AP increase, from 35.3552 to 39.1214, and lowered ECE in every comparison, but it regressed on one TruckScenes seed.

Neither route was better on every criterion. The strict route was more consistent in matched AP, but its ECE increased in all 12 comparisons and it added the cost of expert inference and local voting. The high-performance route was less consistent across seeds, although it gave stronger average accuracy and calibration. These findings are confined to radar-only car detection with four dataset conversions, three seeds, and a PointPillars-style backbone. Further evaluation should choose thresholds on a development set and reproduce recent methods under the same evaluator. Tests on other detectors and classes, together with a faster voting implementation, would show how well the findings carry beyond the current setting.

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Informed Consent Statement: Not applicable.

Data Availability Statement: The paper uses four radar datasets and frozen experimental outputs already recorded in the project workspace. Astyx, MAN TruckScenes, V2X-Radar-V, and K-Radar are available from their original providers subject to their respective access and license terms. Derived experimental configurations, result summaries, manuscript sources, and reproducibility materials supporting the findings are publicly available at the GitHub repository github.com/niubisile-wq/sparse-radar-proposal-reliability and archived at Zenodo (DOI: 10.5281/zenodo.21696782). The original datasets remain subject to the access and licence terms of their providers; raw datasets and model checkpoints are not redistributed in the repository.

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